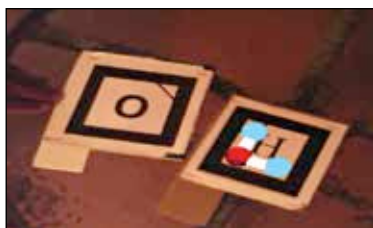


the print. Once the recognition is complete, it replaces the marker on screen with a 3D version of the corresponding object. This allows the user to investigate the object in more detail and from various angles. Rotating the marker would rotate the 3D imagery as well.

- 2 Yet another famous use of recognition AR tech is - translation of words on the fly. In this case, the app reads the words seen by the camera and tries to recognize the words using OCR (Optical Character Recognition) technology and then replaces the words on screen with their translated versions. This can be immensely useful for tourists when travelling to places where the locally prevalent language is not known.
- 3 Recognition based AR can also be used in education. Markers of more than two objects are kept together. The app detects the multiple markers and tries to simulate relationships among them. For example, one can use printed cards to represent atoms (in say chemistry class) and based on their mutual distance the AR app can show how a reaction would take place; and that would be just one use of AR in education. Detection of drawings and sketches by more intelligent apps can help teach small children. e.g. a picture of a giraffe be replaced with a living 3D version of a giraffe and children could see how it looks in reality and they could interact with the same on a touchscreen!



Augemented Reality for simulating chemical formulae

- 4 Recognition of printed versions of 3D objects can help create 3D simulations of those objects without having to actually build a physical model. This can be of great aid to people who constantly work with 3D applications such as architects and animators. We will talk about them in later chapters.
- 5 Recognition based AR can be used in projectors to automatically detect a projectable surface and project on only the projectable area. The projection can be made interactive by using dynamic objects in the surroundings to command the projector. This can eventually be used with the projection mapping technique to auto-detect various types of objects and send out projection imagery according to the size, distance and color of the surface on which projection could be done.